

DIRECTIONS *for questions 1 to 4:* Answer these questions on the basis of the information given below.

In an office, there are exactly four types of employees – Analysts, Managers, Executives and Specialists. The monthly salaries of the employees of the four types are ₹25000, ₹70000, ₹100000, and ₹85000, respectively.

The working days in any week are from Monday to Friday, with Saturday and Sunday being holidays. Any employee can take a leave on any day, but he or she can take a leave only for one or more full days (i.e., leaves of less than one day are not allowed). Further, as per the office policy, an *effective number of leaves* is calculated based on the following criteria:

- If any employee takes a leave on a Friday and on the subsequent Monday, then the *effective number of leaves* for that employee for these days is counted as four (the Friday and the Monday along with the Saturday and Sunday in between).
- In all other cases, the *effective number of leaves* that any employee takes is the same as total number of working days that the employee does not come to the office.

For any employee, the salary for any calendar month is paid on the 10th of the subsequent month, after taking into consideration his *effective number of leaves* during that month. In any calendar month, if the total *effective number of leaves* of the employee during the month (calculated as per the above criteria) is l and the total number of days in the month is n , the salary actually paid (in ₹) to any employee is calculated as follows (rounded off to the nearest integer):

$$Salary\ Paid = \frac{n - l}{n} \times Monthly\ Salary$$

Seven employees, Aniket, Banik, Chunky, Dory, Ekant, Fakir and Gorak, work in the office. The table given below provides, for the months of November and December of a particular year, all the dates on which each employee has taken leave during the working days of the office (i.e., not including Saturdays and Sundays). There were no other holidays during the two months, apart from the Saturdays and the Sundays. The table also provides the type of employee for each employee.

Employee	Type	Leaves in the month of	
		November	December
Aniket	Analyst	1 st , 5 th , 8 th , 11 th , 12 th	5 th , 6 th , 23 rd
Banik	Manager	1 st , 15 th	5 th , 20 th , 26 th , 27 th
Chunky	Analyst	15 th , 18 th , 25 th	9 th , 27 th
Dory	Executive	15 th , 22 nd	23 rd , 26 th
Ekant	Specialist	8 th , 11 th , 22 nd , 25 th	23 rd , 26 th , 30 th
Fakir	Specialist	1 st , 12 th , 15 th	13 th , 16 th
Gorak	Executive	1 st , 5 th , 8 th , 29 th	2 nd , 5 th , 6 th , 9 th

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

What is the maximum total *effective number of leaves* for any employee across the two months combined?

Q2. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

What is the total salary (in ₹) paid to all the employees for the month of November?

☐ a) 4,20,832

☐ b) 4,24,586

☐ c) 4,26,499

☐ d) 4,27,333

Q3. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

What best can be said about the sum of the *effective number of leaves* for the seven employees in December?

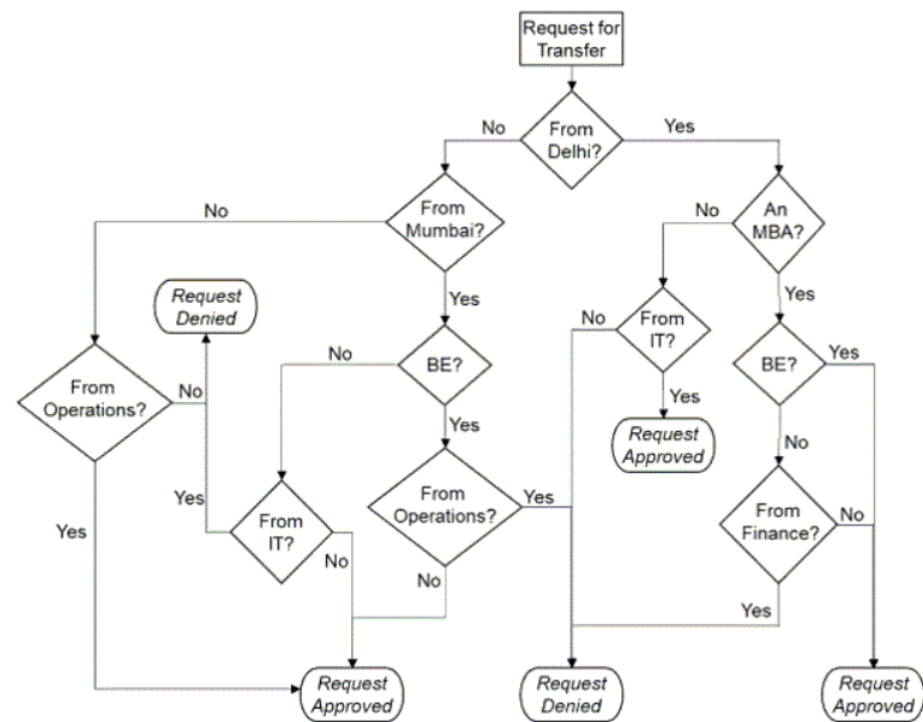
- ☐ a) It is less than 25 but greater than 22.
- ☐ b) It is less than 27 but greater than 24.
- ☐ c) It is less than 29 but greater than 26.
- ☐ d) It is less than 23 but greater than 20.

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

How many employees were paid at most 80% of their monthly salary in at least one of the two months?

DIRECTIONS for questions 5 to 10: Answer these questions on the basis of the information given below.

In an organization, a request for transfer by any employee is approved or denied based on the flow chart given below. The request is either approved or denied based on the location, educational qualification and the department of the employee. Any employee in the organization is from Mumbai or Delhi or Others (representing any other city); he/she has an educational qualification of at least one among MBA or BE; he/she is from exactly one of the departments among Operations, Finance or IT.



Q5. DIRECTIONS *for questions 5 to 7:* Select the correct alternative from the given choices.

The request for transfer of an employee with an MBA and working in Operations department was denied. Which of the following is definitely true of the employee?

- ☐ a) He is from Delhi.
- ☐ b) He does not have a BE.
- ☐ c) He is from Mumbai.
- ☐ d) He is neither from Delhi nor from Mumbai.

Q6. DIRECTIONS *for questions 5 to 7:* Select the correct alternative from the given choices.

The request for transfer of an employee from Mumbai was denied. Which of the following can BEST be said about the employee?

- ☐ a) If he is from the IT department, he must have a BE.
- ☐ b) If he has an MBA, he must be from IT.
- ☐ c) If he has a BE, he must be from the IT department.
- ☐ d) If he is from the Operations department, he must have a BE.

Q7. DIRECTIONS *for questions 5 to 7:* Select the correct alternative from the given choices.

The request for transfer of an employee without BE was approved. Which of the following is definitely false about the employee?

- ☐ a) He works in the Finance department.
- ☐ b) He works in the Operations department and is from Mumbai.
- ☐ c) He works in the IT department and is not from Delhi.
- ☐ d) He is from Delhi.

Q8. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

During a month, there were exactly 180 requests from the employees of each of the three departments. In each department, there were an equal number of requests from employees from Delhi, Mumbai and Others. For each department and city combination, an equal number of requests were from employees who had only MBA, only BE and both.

What is the total number of requests denied during the given month?

Q9. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

During a month, there were exactly 180 requests from the employees of each of the three departments. In each department, there were an equal number of requests from employees from Delhi, Mumbai and Others. For each department and city combination, an equal number of requests were from employees who had only MBA, only BE and both.

Among the requests from employees with an MBA, how many requests were accepted?

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

During a month, there were exactly 180 requests from the employees of each of the three departments. In each department, there were an equal number of requests from employees from Delhi, Mumbai and Others. For each department and city combination, an equal number of requests were from employees who had only MBA, only BE and both.

Among all the requests that were rejected, the maximum number of requests were from employees of which department?

- ☐ a) Finance
- ☐ b) IT
- ☐ c) Operations
- ☐ d) Both Operations and IT

DIRECTIONS *for questions 11 to 14:* Answer these questions on the basis of the information given below.

King John built a monument to honour his father, King Jack. The monument comprised seven pillars of different heights, in a row, from left to right. He built the monument along a road, so that the seven pillars can be viewed from left to right by any person when they stand on the road in front of the monument.

The heights of the seven pillars signified the number of wars that King Jack waged during different years of his reign, and the heights were 10 ft, 30 ft, 45 ft, 50 ft, 75 ft, 80 ft and 85 ft, not necessarily in the same order. Further, the pillars were of three different colours among Red, Blue and Green, as these were the colours present in his father's coat of arms.

It is also known that

- i. no two pillars adjacent to each other were of the same colour and there were at least two pillars of each of the three colours.
- ii. the sum of the heights of all the Red coloured pillars was 10 ft more than the total height of the first three pillars from the left.
- iii. the shortest pillar is to the immediate left of the tallest pillar, and neither are Blue in colour.
- iv. the second pillar from the left is 30 ft higher than the second pillar from the right, and neither are Red in colour.
- v. the fifth pillar from the left is Red in colour, while the pillar at the extreme right, which is not Blue in colour, is 20 ft shorter than the pillar at the extreme left.

Q11. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

What is the height (in ft) of the third pillar from the left?

- ☐ a) 10
- ☐ b) 75
- ☐ c) 80
- ☐ d) 50

Q12. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

What BEST can be said about the total height of all the pillars that are Blue?

- ☐ a) Exactly 120 ft
- ☐ b) Either 55 ft or 120 ft
- ☐ c) Either 85 ft or 120 ft
- ☐ d) Exactly 85 ft

Q13. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

What BEST can be said about the colour of the second pillar from the left?

- ☐ a) Blue
- ☐ b) Green
- ☐ c) Either Green or Red
- ☐ d) Either Blue or Green

Q14. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

What is the maximum difference (in ft) in the heights of any pair of pillars of the same colour?

☐ a) 30

☐ b) 35

☐ c) 50

☐ d) 45

DIRECTIONS *for questions 15 to 20:* Answer these questions on the basis of the information given below.

Eight students, A through H, were standing in a queue, one in front of the other, not necessarily in the same order. Each of the eight students had either one or two marbles with him. Further, all the marbles that each student had were of the same colour, which was either blue or red.

Kiran, the teacher for these students, gave a certain number of green marbles to each of the eight students based on the initial number of marbles that the student had, in the following manner:

- If any student had the same number and the same colour of marbles as the student standing immediately behind him, Kiran gave him three green marbles.
- If any student had the same number but not the same colour of marbles as the student standing immediately behind him, Kiran gave him two green marbles.
- If any student had the same colour but not the same number of marbles as the student standing immediately behind him, Kiran gave him one green marble.
- In all the other cases (including for the student standing at the end of the queue), Kiran gave the student four green marbles.

After Kiran gave the marbles to all the students, the number of marbles with the eight persons, A through H were 2, 4, 5, 2, 5, 4, 6 and 3, respectively.

The following information is also known about the students in the queue, after Kiran gave the marbles:

- i. the total number of marbles with the students in the fourth and seventh places was 7.
- ii. H was not standing immediately behind B, and Kiran gave exactly 2 marbles to B, who was the third in the queue.
- iii. G was standing immediately in front of D, while B had at least one red marble with him.
- iv. E was neither standing next to H nor was E in the fourth position.

Q15. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

What BEST can be said about the total number of marbles that Kiran gave?

- ☐ a) Exactly 17
- ☐ b) Exactly 18
- ☐ c) Exactly 19
- ☐ d) Either 18 or 19

Q16. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

Which of the following BEST describes the marbles that E had with him initially?

- ☐ a) Two red marbles
- ☐ b) One blue marble
- ☐ c) Two blue marbles
- ☐ d) One red marble

Q17. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

Who is standing immediately in front of H?

☐ a) C

☐ b) B

☐ c) D

☐ d) A

Q18. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

Which of the following can be the marbles that C had with him after Kiran gave him the green marbles?

- I. One red marble and four green marbles

- II. Two red marbles and three green marbles

- III. One blue marble and four green marbles

- IV. Two blue marbles and four green marbles

- ☐ a) II or III

- ☐ b) I or II or III

- ☐ c) Only III

- ☐ d) II or IV

Q19. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

What is the total number of marbles with the first four persons in the queue before Kiran gave them the green marbles?

Q20. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

What is the difference between the total number of red marbles and the total number of blue marbles with all the eight students combined?